

A Finite-Horizon Guidewire Education Model for Minimally Invasive Total Knee Arthroplasty Training

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Abstract

We present a Nature-style preprint for the new knee surgery education tab added to shsc_vr_genai. The module couples guidewire diameter, stiffness, insertion-angle bias, minimally invasive exposure, and coalition training mission profiles over a finite horizon of 36 depth samples. In the dashboard reference configuration, the simulator projects a canal capture probability of 88.7%, an alignment index of 78.2%, a workflow readiness score of 83.8%, and a digital interoperability score of 86.0%. These outputs support minimally invasive total knee arthroplasty rehearsal, structured workflow coaching, and readiness analytics for US and Canadian military education programs.

Finite-Horizon Formulation

All dashboard quantities are evaluated on a finite depth lattice $k = 0, \dots, N$ with $N = 35$ increments spanning 0 to 120 mm. The formulation is algebraic rather than continuous so the preprint mirrors exactly what is rendered inside the education tab and can be audited by instructor teams.

$$e_k = 0.28 + \frac{1.35 \eta_a \eta_\theta k}{\eta_s \eta_d N} + 0.06 \sin\left(\frac{3\pi k}{N}\right), \quad k = 0, \dots, N$$

$$\ell_k = 15 + 0.18S + 4.8D + 16\eta_a \frac{k}{N} + 1.2|\theta|$$

$$R = 0.45A + 0.35C + 0.20P, \quad I_m = b_m + 0.12(A - 80) + 0.08(C - 80) - 0.35F$$

Here, D is guidewire diameter in millimetres, S is the stiffness index, θ is insertion-angle bias in degrees, η_a is the minimally invasive exposure factor, $\eta_\theta = 1 + |\theta| / 7$, η_s is the stiffness gain, η_d is the diameter gain, A is the alignment index, C is canal capture probability, P is the soft-tissue preservation score, F is the number of fluoroscopy verification passes, and b_m is the mission-specific digital transformation baseline.

Reference Scenario and Educational Meaning

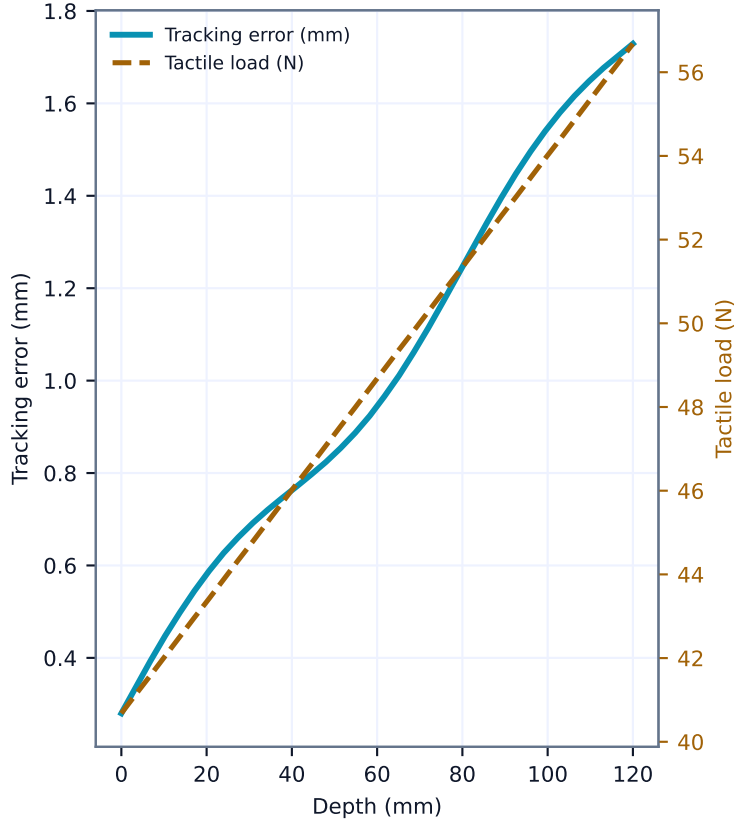
The reference tab configuration uses the Mini-midvastus MIS TKA pathway with a 2.8 mm guidewire, stiffness 58, and 1.5 degree angular bias for the joint US/Canadian allied initiative. An optimized subvastus rehearsal scenario at 3.0 mm, stiffness 64, and 0.6 degree angular bias raises canal capture to 94.6% and alignment to 82.7%, illustrating that angular control and moderate stiffness gains are the dominant levers for finite-horizon improvement.

US / Canadian Allied Initiative uses the Mini-midvastus MIS TKA pathway to pair minimally invasive exposure discipline with repeatable guidewire checkpoints. The current configuration projects a canal capture probability of 88.7% and an implant alignment index of 78.2%. Digital transformation focus: Standardized workflow guides, cloud-hosted simulation evidence, and interoperable after-action analytics.

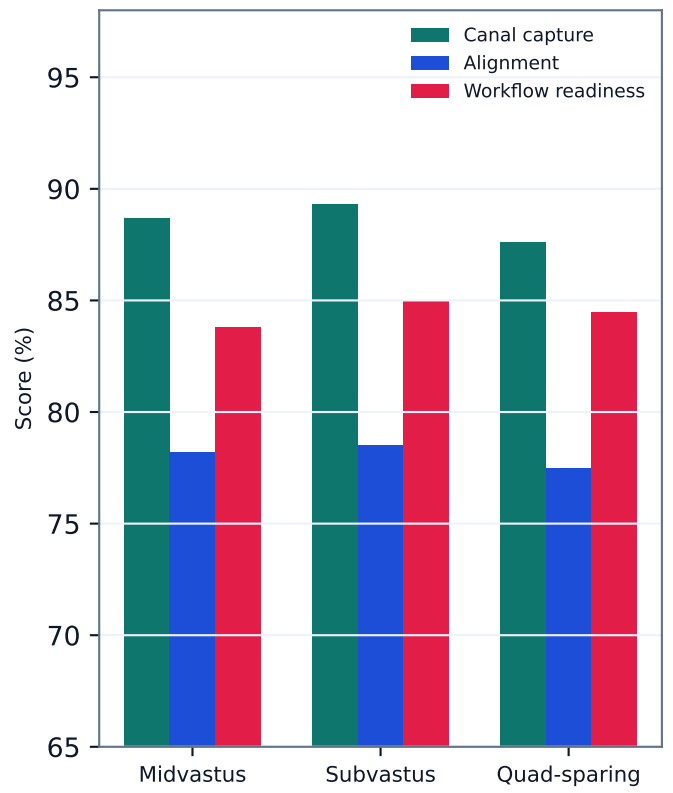
Finite-Horizon Knee Education Simulation Outputs

Reference scenario: Mini-midvastus | Joint allied initiative | 2.8 mm wire | stiffness 58 | angle 1.5 deg

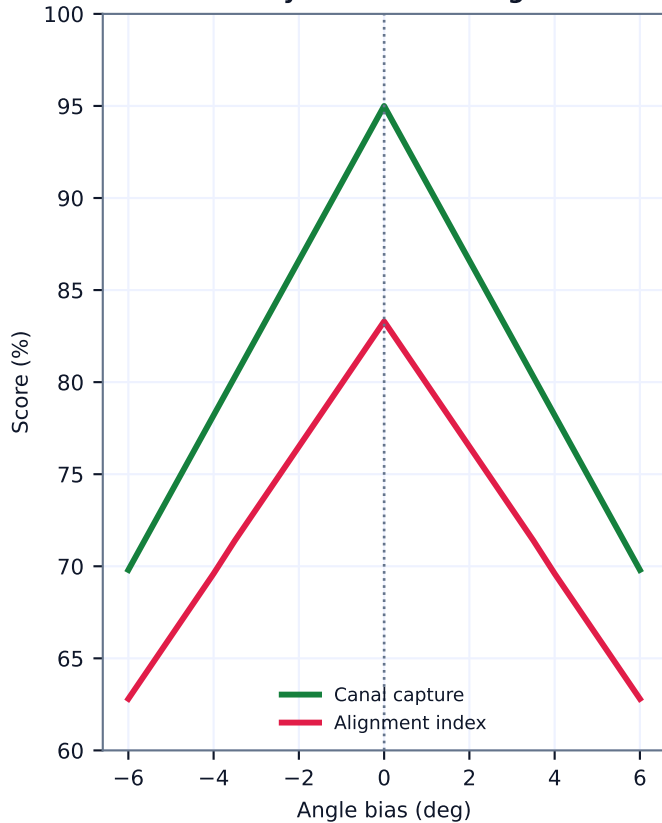
Guidewire tracking and tactile load vs depth



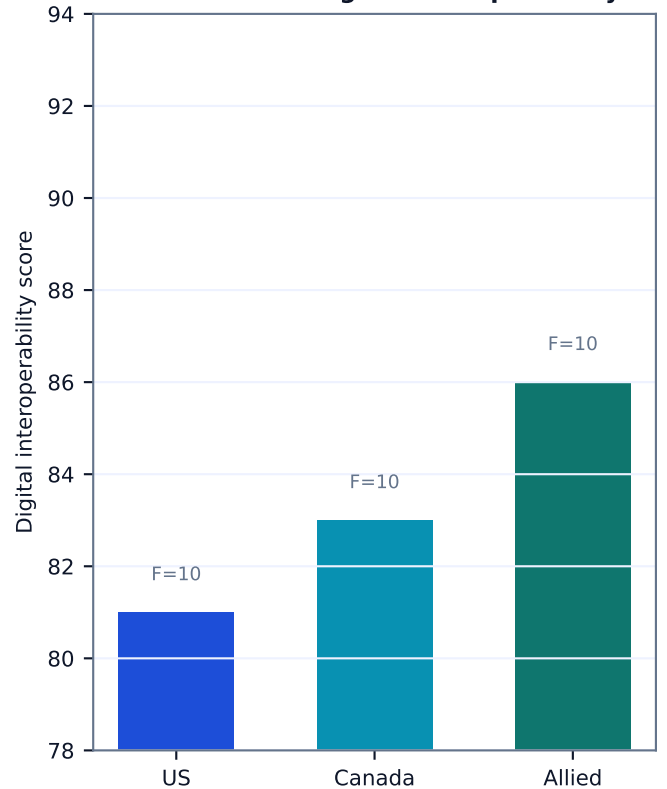
Approach comparison at fixed wire settings



Sensitivity to insertion-angle bias



Mission-level digital interoperability



Finite workflow ladder from the tab



Operational interpretation

The knee education tab is most useful when treated as a finite rehearsal system rather than a free-form simulator. The guidewire equations define a constrained space in which angular bias drives both canal capture and downstream implant alignment. The workflow layer then translates those continuous-looking curves into five discrete coaching phases that an instructor can evaluate during a single session.

US and Canadian military digital transformation framing

US Military Medical Education: Distributed readiness pipeline for expeditionary orthopedic teams. Priority: Common metrics, simulation-backed sustainment, and faster remediation for deployed care teams. Model score = 81.0. Canadian Armed Forces Education: Joint digital surgery curriculum supporting austere and coalition care settings. Priority: Interoperable training records, bilingual-ready content pathways, and team-based performance review. Model score = 83.0. US / Canadian Allied Initiative: Shared digital transformation track for bilateral orthopedic simulation and competency reporting. Priority: Standardized workflow guides, cloud-hosted simulation evidence, and interoperable after-action analytics. Model score = 86.0.

Conclusion

This preprint translates the knee surgery education tab into a publication-ready model with explicit finite equations, figure outputs, and workflow guidance. The resulting PDF can support instructor calibration, program briefs, and internal digital transformation proposals without over-claiming clinical evidence. Its main value is educational traceability: the same simulation logic now drives the dashboard, the explanatory mathematics, and the exported preprint artifact.

Structured minimally invasive workflow guide

1. Pre-op digital templating

Objective: Review CT or long-leg radiographs, confirm implant sizing, and pre-brief instrument flow. Teaching focus: Align distributed planning artifacts across simulation stations and OR teams.

2. Portal and arthrotomy setup

Objective: Establish a muscle-sparing midvastus window with direct visualization of the trochlear entry corridor. Teaching focus: Maintain extensor continuity while preserving a reproducible exposure window for trainees.

3. Guidewire and canal control

Objective: Advance the guidewire on the intended femoral or tibial axis and confirm capture before reaming or cutting. Teaching focus: Translate wire feel, fluoroscopy cues, and tracking overlays into a stable trajectory decision.

4. Bone preparation and balancing

Objective: Perform measured resections, gap balancing, and trial verification through the minimized exposure. Teaching focus: Teach sequencing discipline so limited visualization does not degrade alignment checks.

5. Implantation and closure

Objective: Seat components, verify patellar tracking, and close with recovery-oriented soft-tissue handling. Teaching focus: Use structured after-action review metrics for readiness reporting.